

Report_DEB

1. Importation des données

```
file_path <- here::here("mod/Correction_X/B_DEB_SAFLYe")

l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  #"Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
  "Bart2020",
  #"Gollot2026_lufa"
  "Gollot2026_dens_D1",
  "Gollot2026_dens_D2",
  "Gollot2026_dens_D2b",
  "Gollot2026_dens_D5",
  "Gollot2026_dens_D7"
)

Adults_alone <- FALSE

df_data <- f_import_data_DEB(l_Experiments, Adults_alone)
```

2. Model fit

```

NbIter <- 100000
#seeds <- c(C1 = 3336, C2 = 6663, C3 = 1222, C4 = 2420, C5 = 4848, C6 = 5656, C7 = 7077, C8 = 8484)
seeds <- c(C1 = 3303, C2 = 3666, C3 = 2212)

text_priors <- '
Distrib (Fm,          TruncNormal,          0.1,      0.2, 0.0001,      0.8);
Distrib (pAm,         TruncNormal_cv,        100,      0.2, 40,      2000);
Distrib (r_pAm_pM,    TruncNormal_cv,        0.8,      0.1, 0.1,      2);
Distrib (v,           TruncNormal_cv,        0.18505,   0.1, 0.001,    2);
Distrib (kap,         Uniform,                0,        1);
Distrib (kapA,        Uniform,                0.1,      0.9);
Distrib (Ehp,         TruncNormal_cv,        100,      0.2, 40,      300);
Distrib (E_coc,       TruncNormal_cv,        30,      0.2, 1,      300);
Distrib (rOM_ClxHorse, LogUniform,          0.00001,   0.3);
Distrib (dv,          TruncNormal_cv,        2,      0.5, 0,      10);
Distrib (wE,          LogUniform,          0.000000001, 0.1);

# Calcul des vraisemblances
Distrib (Sigma_W, TruncNormal, 5, 1, 0.01, 100);'

text_likelihood <- 'Likelihood (Weight,          Normal, Prediction(Weight), Sigma_W);
Likelihood (Reproduction, Poisson, Prediction(Reproduction));'

OM_diff <- TRUE
RTOL <- 1e-7
ATOL <- 1e-6

# makemcsim DEBcali.model

#f_In_tot(file_path, l_Experiments, Adults_alone, OM_diff, text_priors, text_likelihood, NbIter)

# ./mcsim.DEBcali DEB_C1.in

```

OM_horse and OM_soil distinct ? TRUE

Adult and alone earthworm taken into account ? FALSE

3. Results

	Fm.1.	pAm.1.	r_pAm_pM.1.	v.1.	kap.1.	kapA.1.
Result_mode	0.2989020	296.63800	0.5248500	0.20434100	0.89700100	0.3989630

2.5%	0.0703333	139.55900	0.3873230	0.16546200	0.56748025	0.1208300
97.5%	0.5556195	329.07200	0.6422690	0.23825160	0.91549495	0.8809430
Min	0.0550587	81.25320	0.2350730	0.12266700	0.44284000	0.1000350
Max	0.7992720	352.32500	0.8408770	0.26977900	0.93844900	0.8998250
Result_mean	0.2749405	283.92348	0.5024032	0.20303842	0.86697668	0.5026899
Result_sd	0.1179708	47.33482	0.0534951	0.01855317	0.07667668	0.2317076
	Ehp.1.	E_coc.1.	rOM_ClxHorse.1.	dv.1.		wE.1.
Result_mode	40.092200	2.1256900	0.0072196600	8.078060	1.012710e-09	
2.5%	40.147415	1.2349400	0.0009353870	1.734020	1.073910e-09	
97.5%	55.782900	3.1182700	0.0138890650	9.030750	6.578323e-07	
Min	40.000100	1.0007400	0.0000974791	0.522040	1.012710e-09	
Max	138.460000	35.8101000	0.2894790000	9.974000	6.832800e-06	
Result_mean	44.810466	2.3374607	0.0068885303	6.751668	1.083194e-07	
Result_sd	4.300476	0.4946532	0.0037284886	1.874521	3.326557e-07	
	Sigma_W.1.					
Result_mode	0.2357890					
2.5%	0.2023080					
97.5%	0.6226430					
Min	0.1781960					
Max	6.5315500					
Result_mean	0.2748964					
Result_sd	0.1229888					

Number of observations, n = 277
 Number of parameters, k = 11
 Log-likelihood, LL = -114.9787
 BIC = 291.8216

Potential scale reduction factors:

	Point est.	Upper C.I.
dv.1.	1.79	3.15
E_coc.1.	1.01	1.02
Ehp.1.	1.05	1.14
Fm.1.	1.00	1.01
kap.1.	1.07	1.22
kapA.1.	1.00	1.02
pAm.1.	1.04	1.11
r_pAm_pM.1.	1.47	2.29
rOM_ClxHorse.1.	1.00	1.00
Sigma_W.1.	1.11	1.32

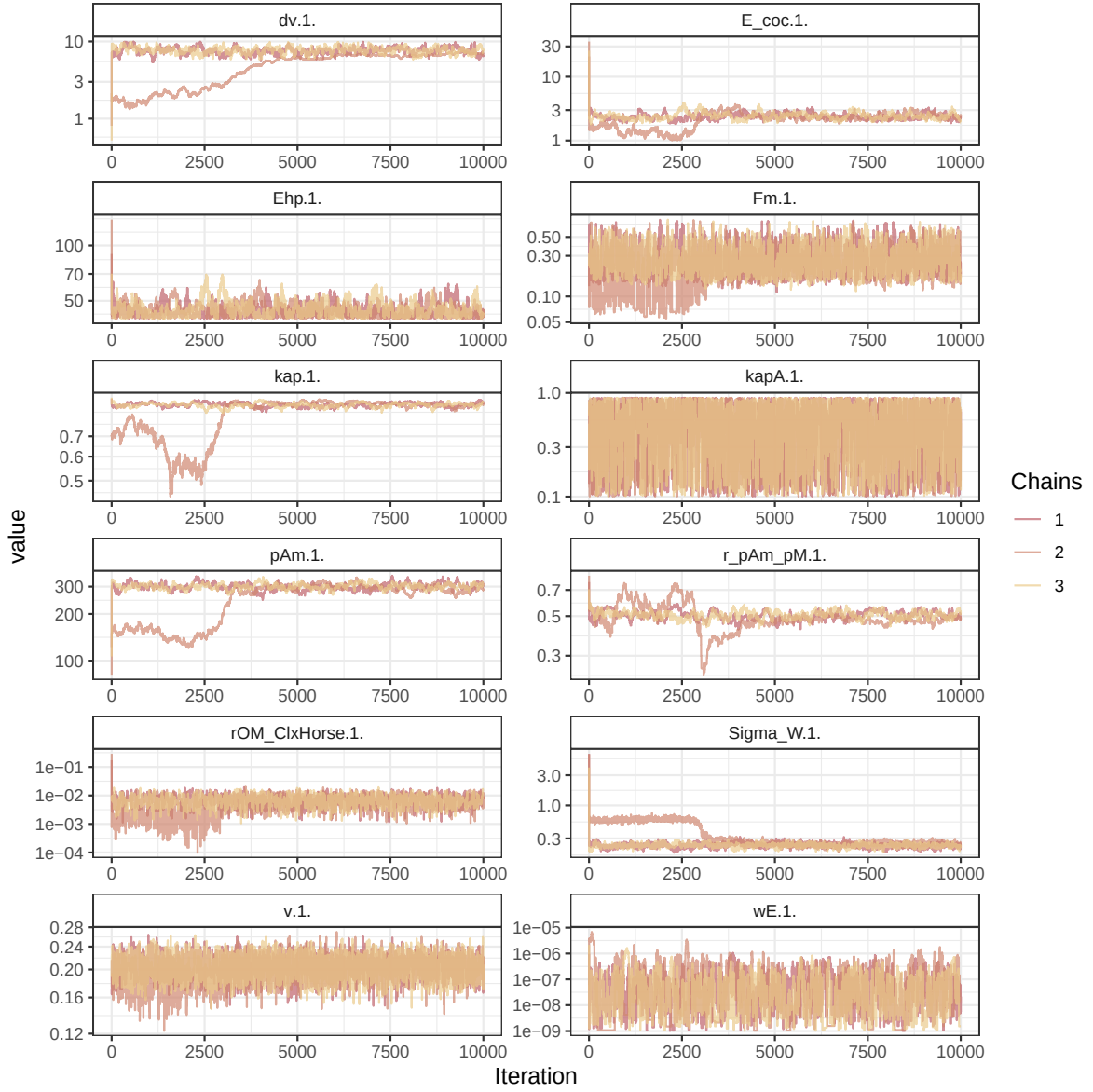


Figure 1: Parameters chains. Only the first iteration and the last `r Nb_iter_kept` are represented.

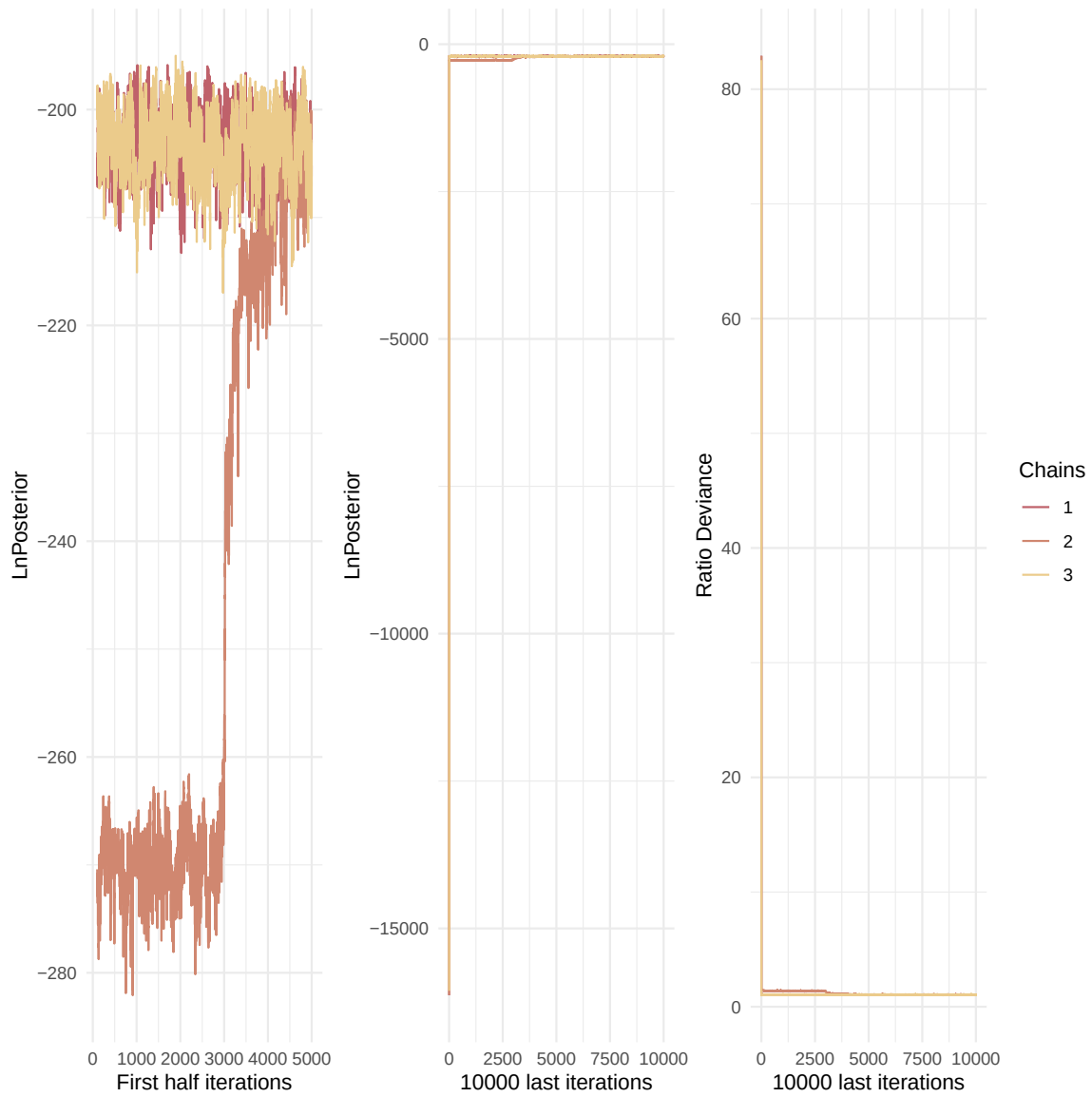


Figure 2: Evolution of likelihood and deviance ratio through the iterations.

v.1.	1.00	1.00
wE.1.	1.03	1.05

Multivariate psrf

1.52

```
mcmc_list_corr <- mcmc.list(  
  lapply(mcmc_list, function(x) {  
    mcmc(as.matrix(x)[-1, , drop = FALSE]) # <- reconvertir en mcmc  
  })  
)  
  
colramp_aurora <- colorRampPalette(rev(Nord_aurora[1:4]))  
ipairs(as.matrix(mcmc_list_corr), colramp = colramp_aurora, pixs=0.25, cex.diag = 0.5)
```

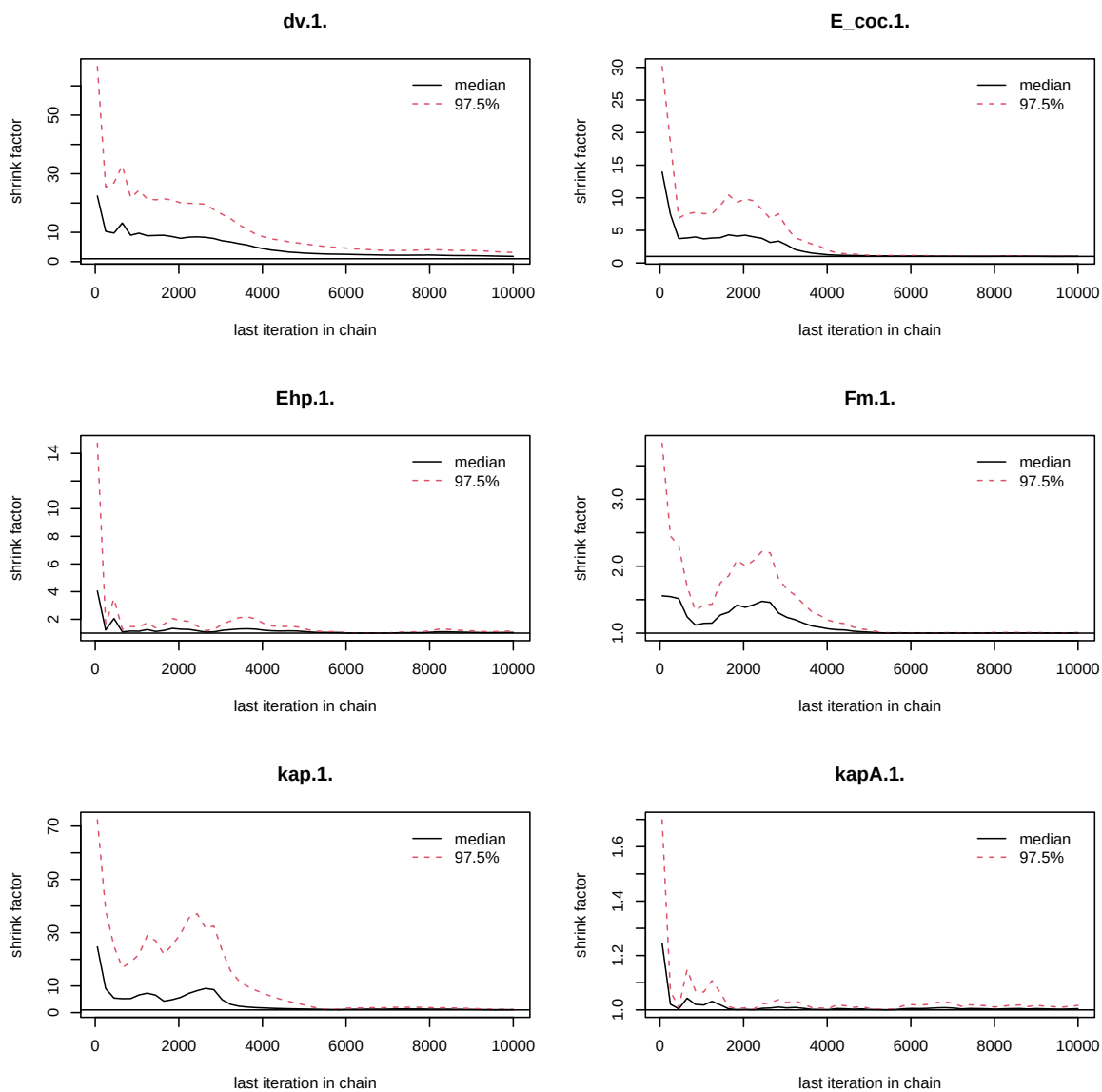


Figure 3: Chains convergence.

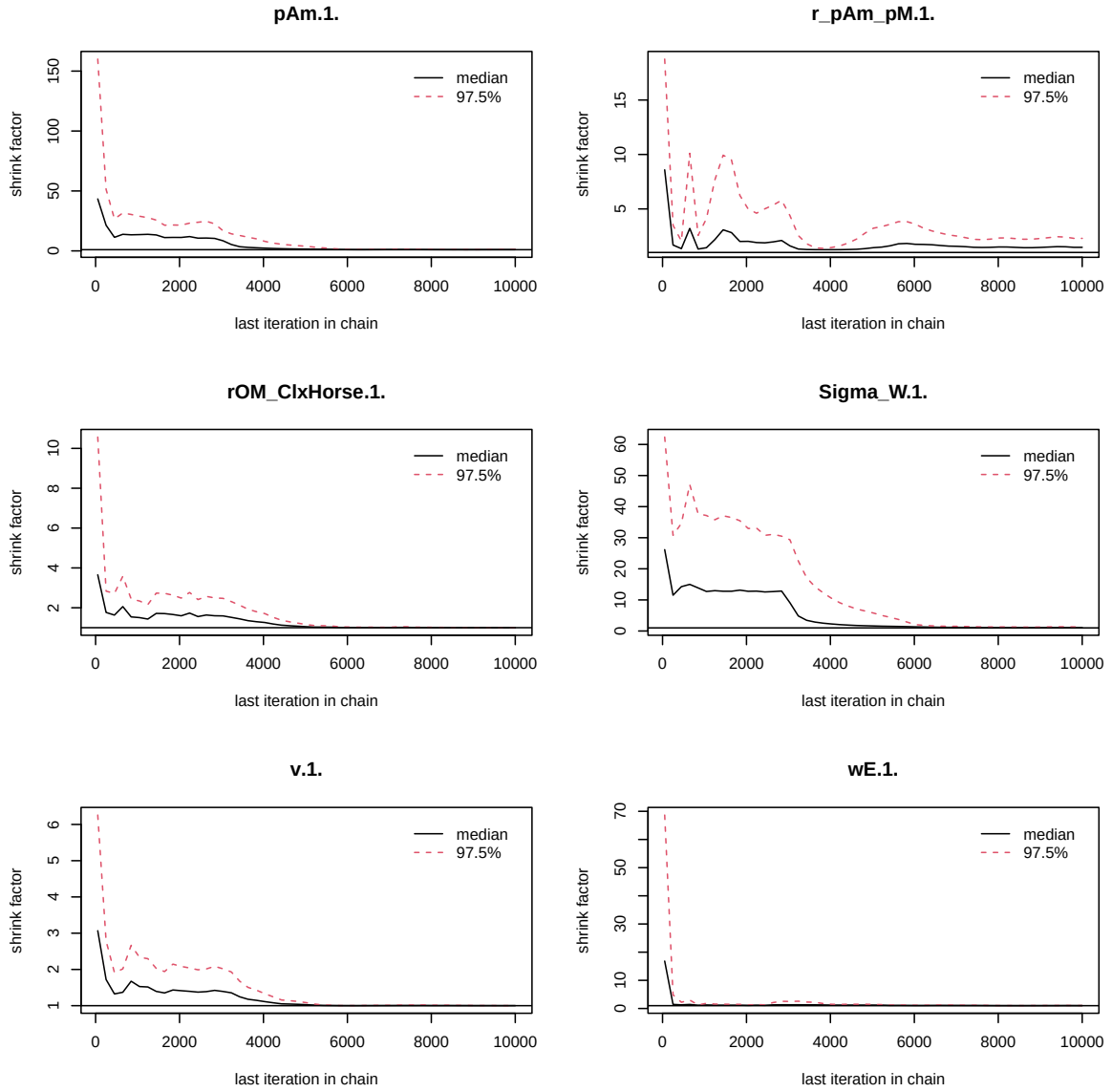


Figure 4: Chains convergence.

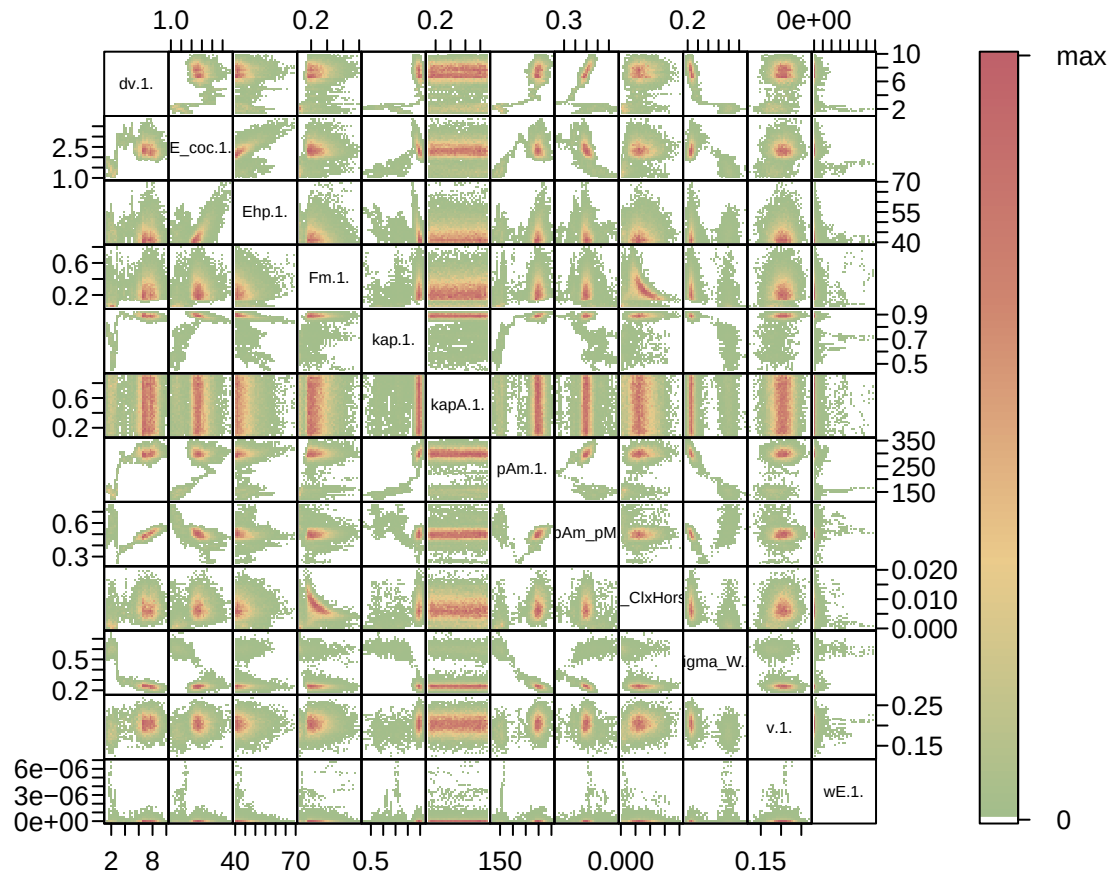


Figure 5: Parameter correlations.

4. Simulations

```
l_Experiments <- c(
  "Bart2019_XP1_1",
  "Bart2019_XP1_2",
  "Bart2019_XP2",
  "Bart2019_XP3",
  "Bart2019_XP4_1",
  "Bart2019_XP4_2",
  "Bart2019_XP5",
```

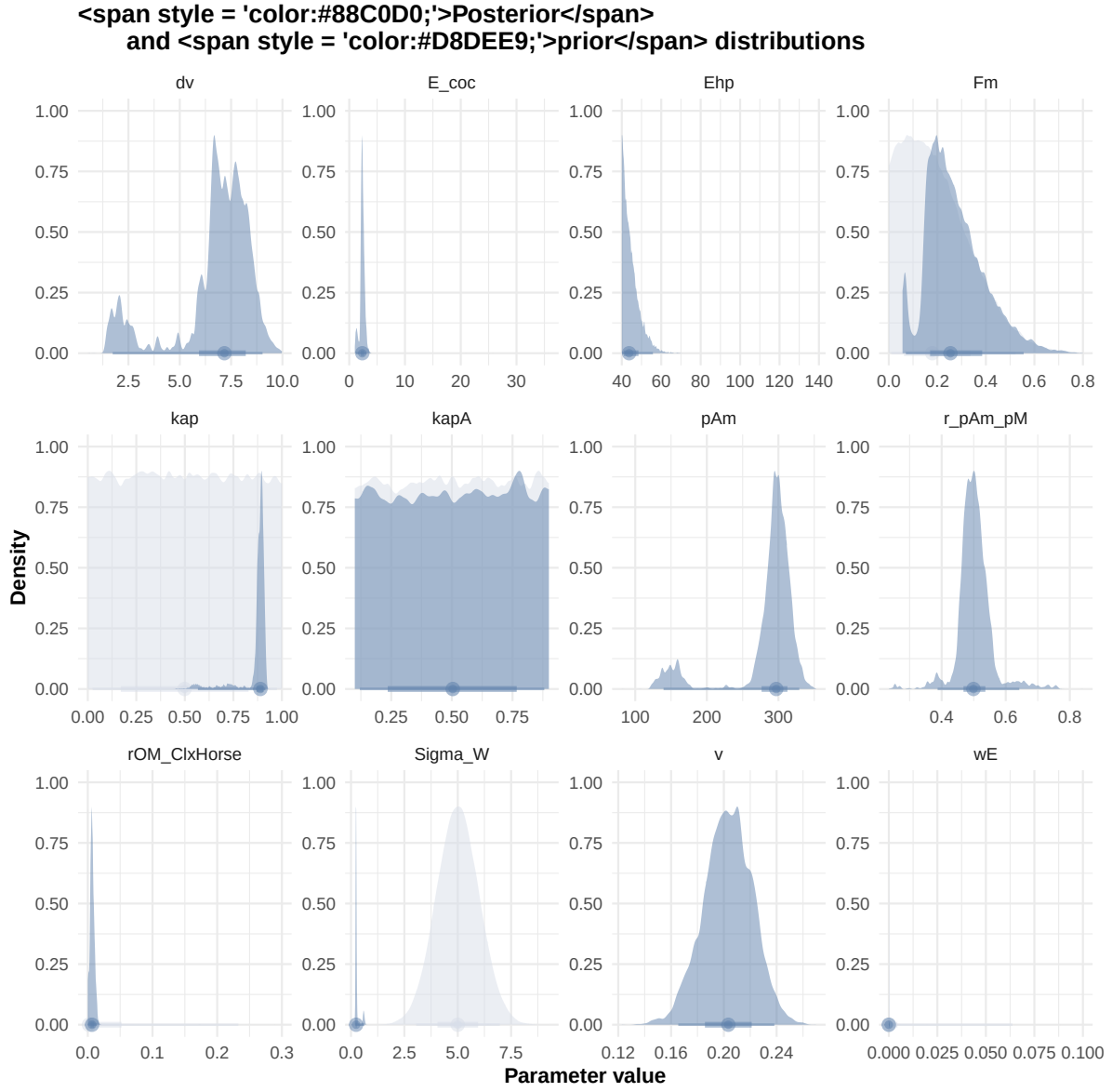


Figure 6: Distributions of the priors and the posteriors.


```
#####
df_No_sim <- df_data |>
  dplyr::select(ID_experiment, No_sim)

df_carac_color <- data.frame(
  carac = unique(l_carac),
  color = c(Nord_aurora[1], Nord_polar[2], Nord_aurora[3], Nord_aurora[2], Nord_aurora[5],
            Nord_aurora[5], Nord_frost[4], Nord_frost[3], Nord_frost[2], Nord_frost[1])
)
v_color_carac <- setNames(df_carac_color$color, df_carac_color$carac)
#####

df_data <- f_import_data_DEB(l_Experiments, Adults_alone) |>
  left_join(df_carc_exp, by="ID_experiment")

# time_limits <- df_data %>%
#   group_by(ID_experiment) %>%
#   summarise(time_cutoff = max(Time, na.rm = TRUE) + 5)

select_times <- seq(0,400, 0.1)

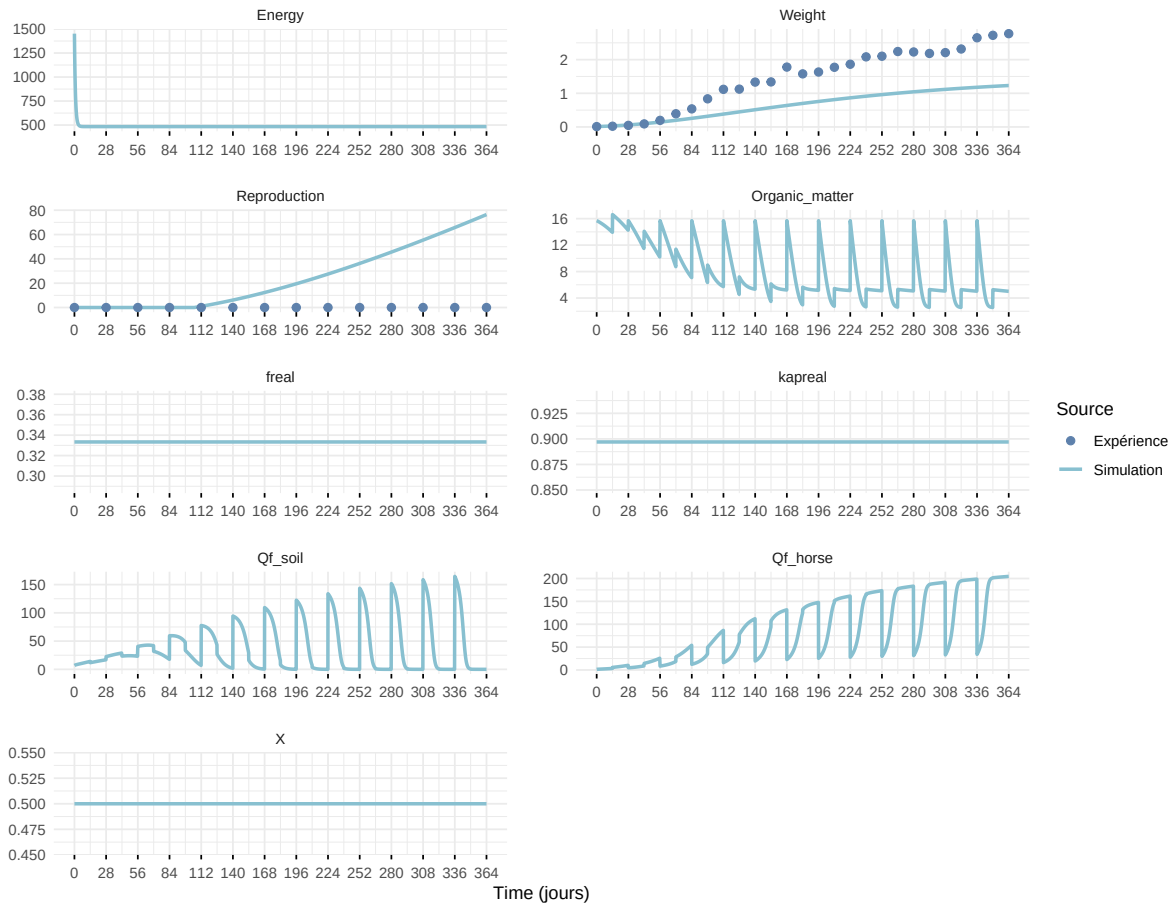
df_sim <- f_MCSim_read_sim(here::here(file_path, "sim.out")) |>
  left_join(df_No_sim, by="No_sim") |>
  left_join(df_carc_exp, by="ID_experiment") |>
  # left_join(time_limits, by = "ID_experiment") |>
  # filter(Time <= time_cutoff) |>
  # dplyr::select(-time_cutoff) %>%
  filter(Time %in% select_times) |>
  mutate(
    Reproduction = case_when(
      Reproduction <= 5e-6 ~ 0,
      .default = Reproduction
    )
  )
)
```

Warning: `read\_table2()` was deprecated in readr 2.0.0.
 i Please use `read\_table()` instead.

Warning in left_join(f_MCSim_read_sim(here::here(file_path, "sim.out")), : Detected an unexpected join
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship =`

"many-to-many" to silence this warning.

Expérience : Bart2019_XP1_1



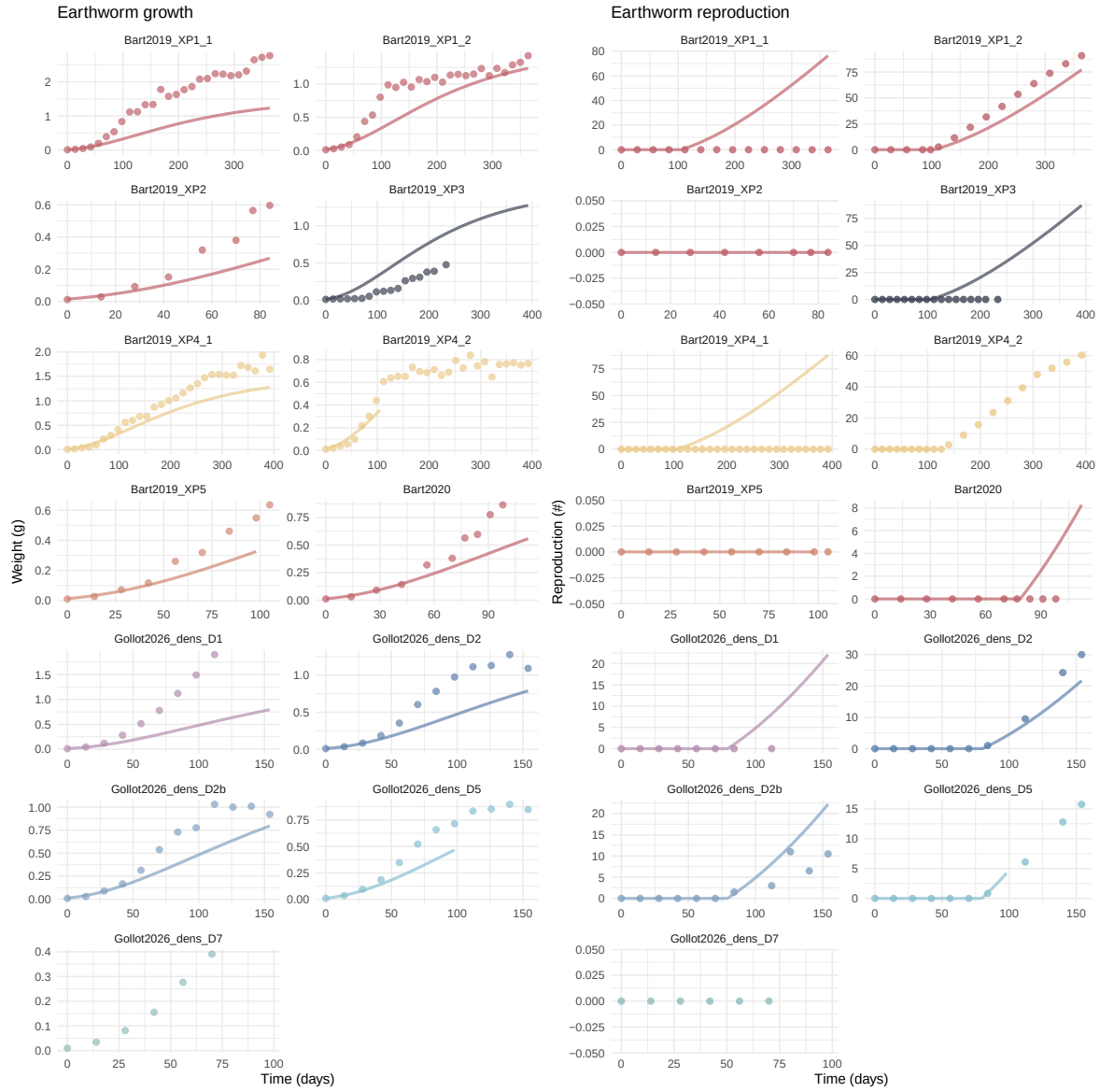
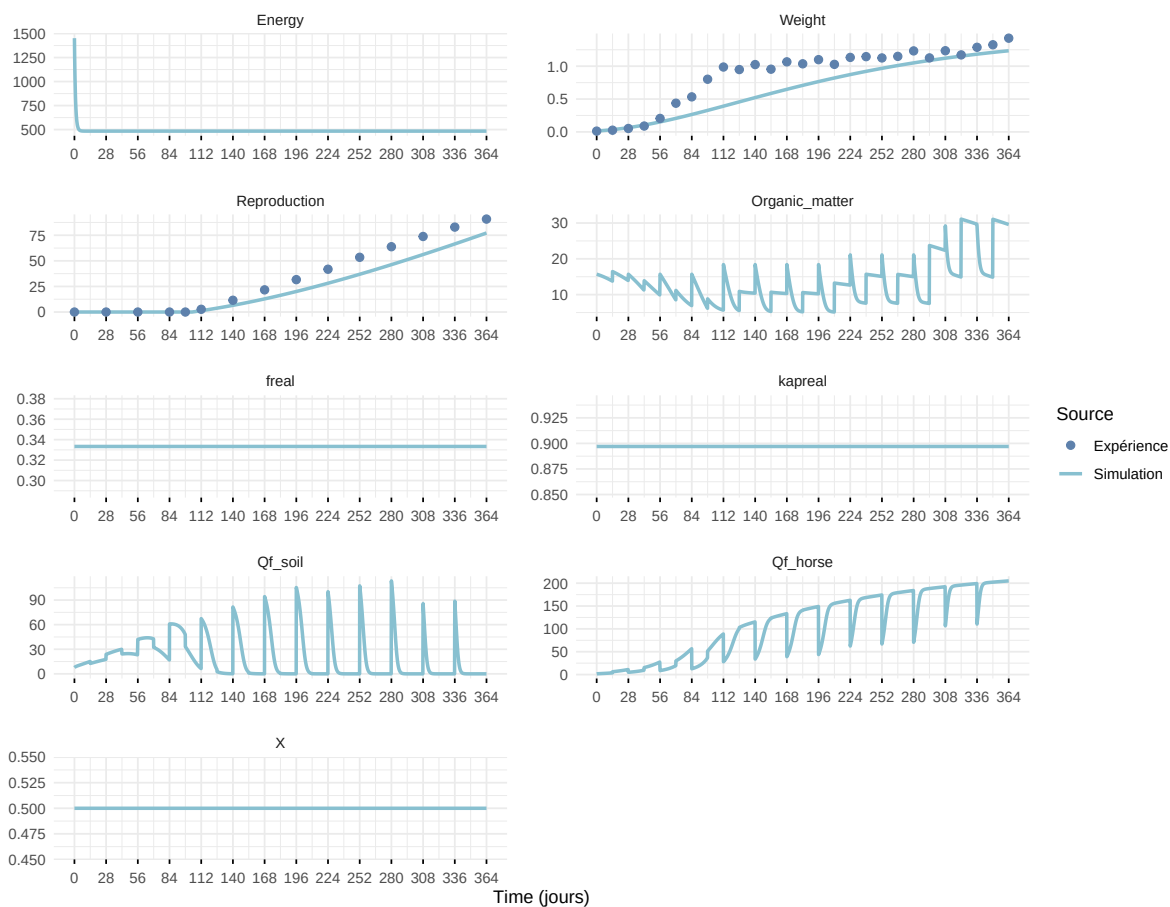
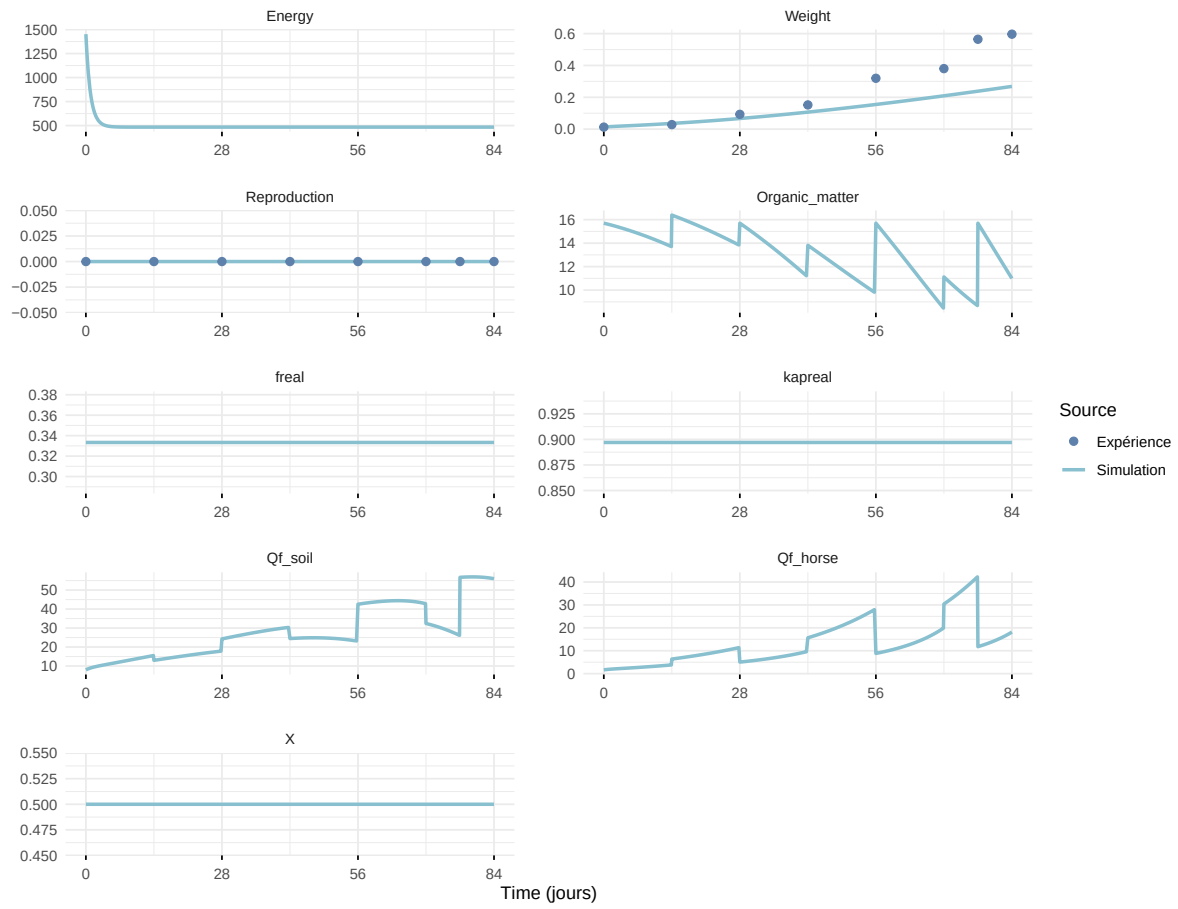


Figure 7: Simulations Weight and Reproduction.

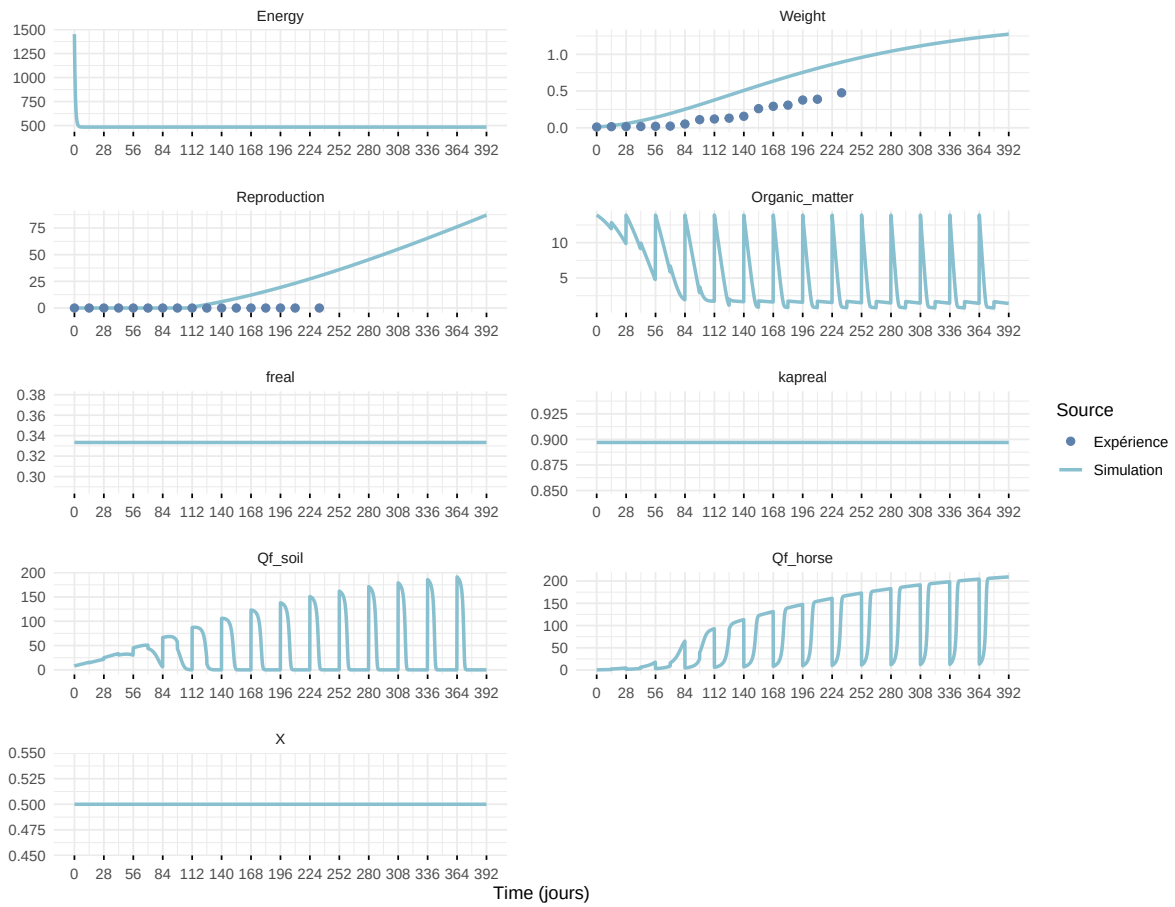
Expérience : Bart2019_XP1_2



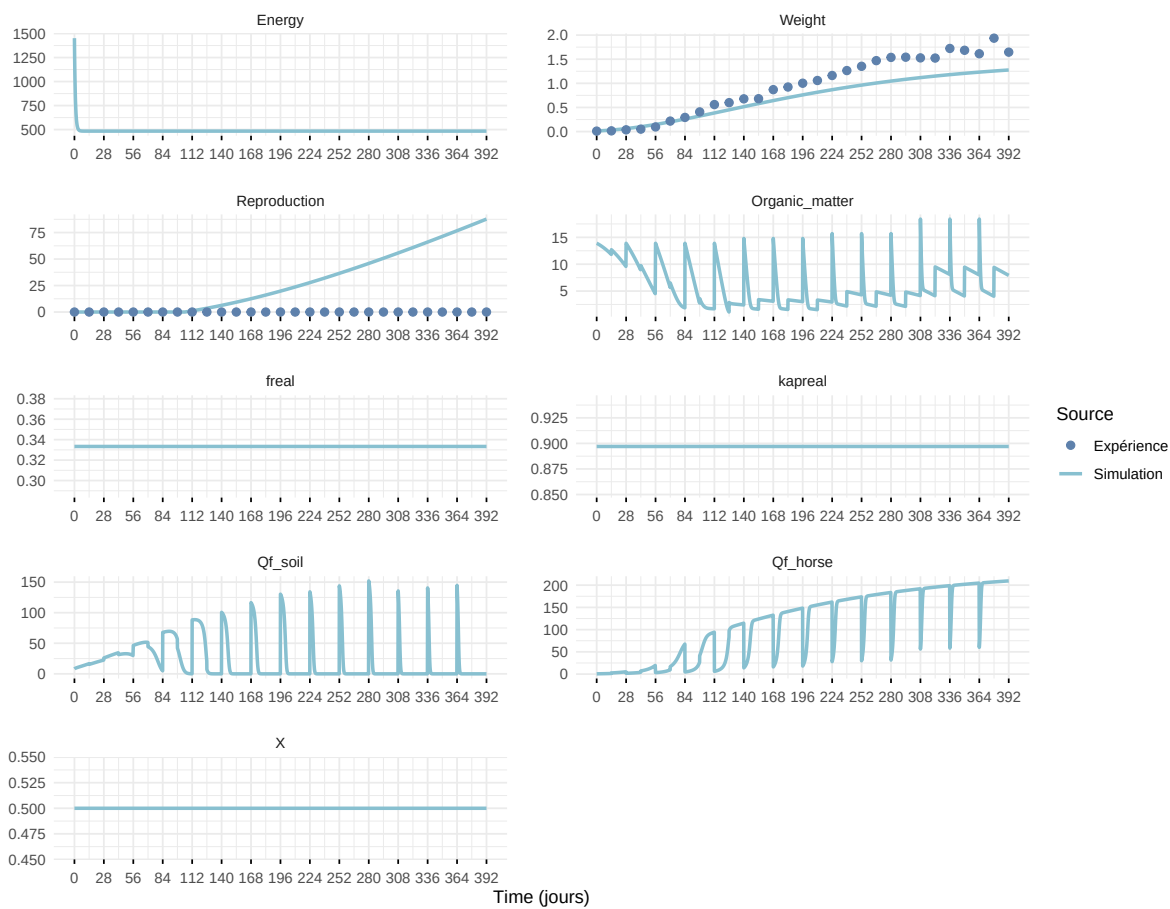
Expérience : Bart2019_XP2



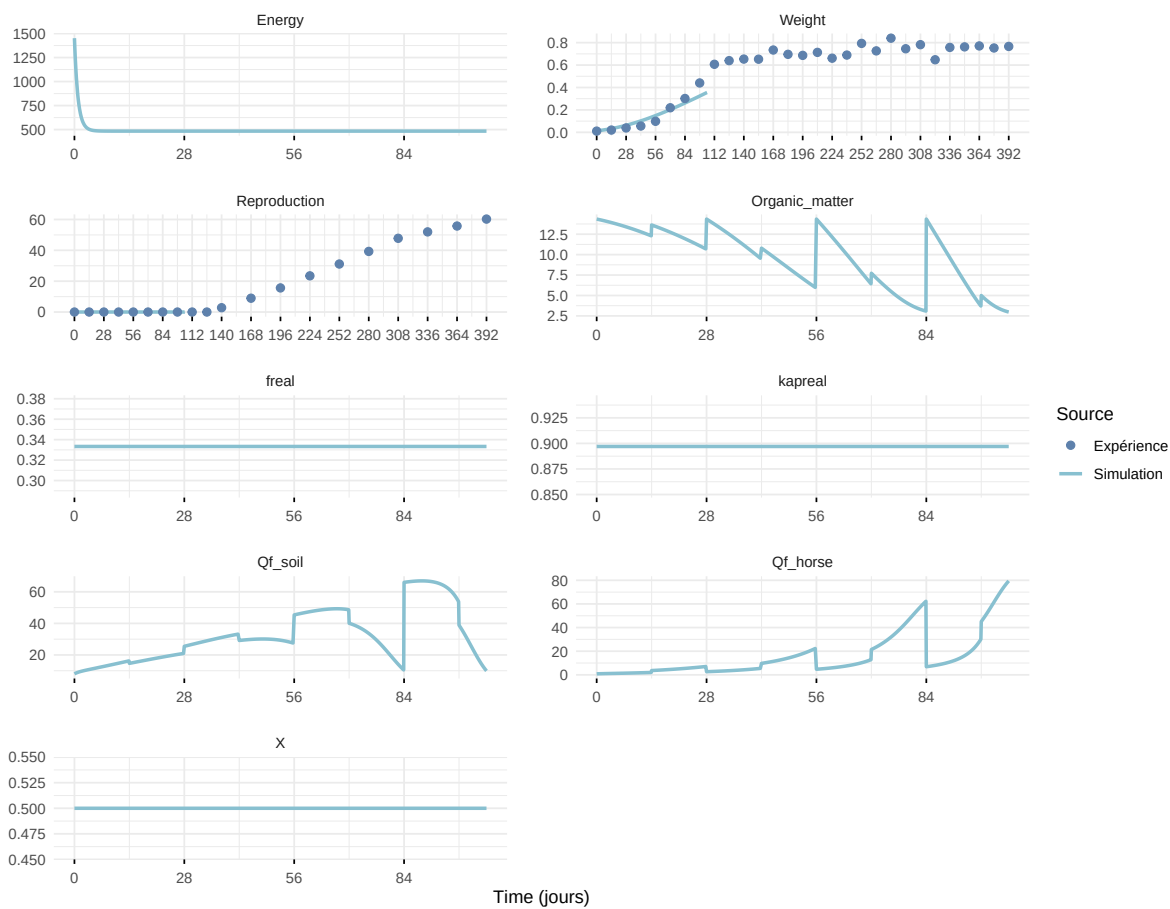
Expérience : Bart2019_XP3



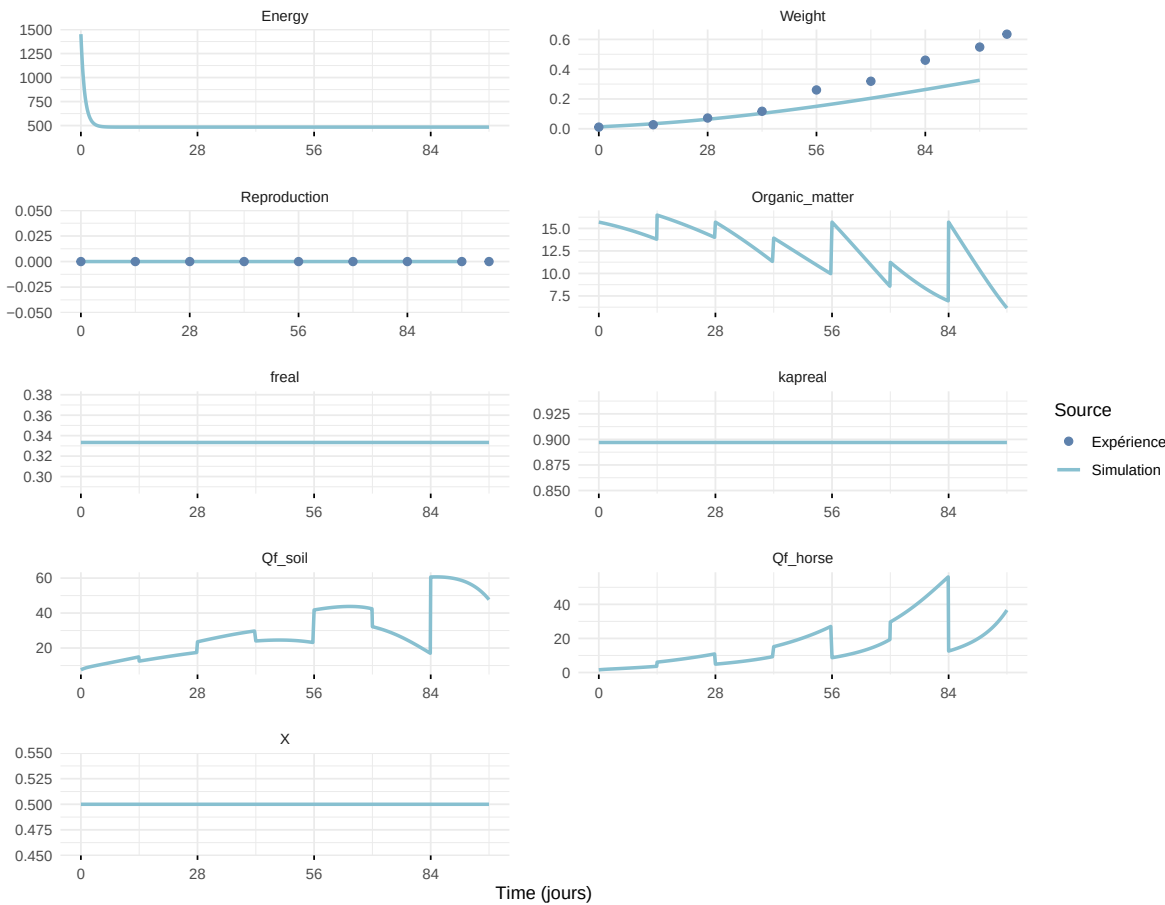
Expérience : Bart2019_XP4_1



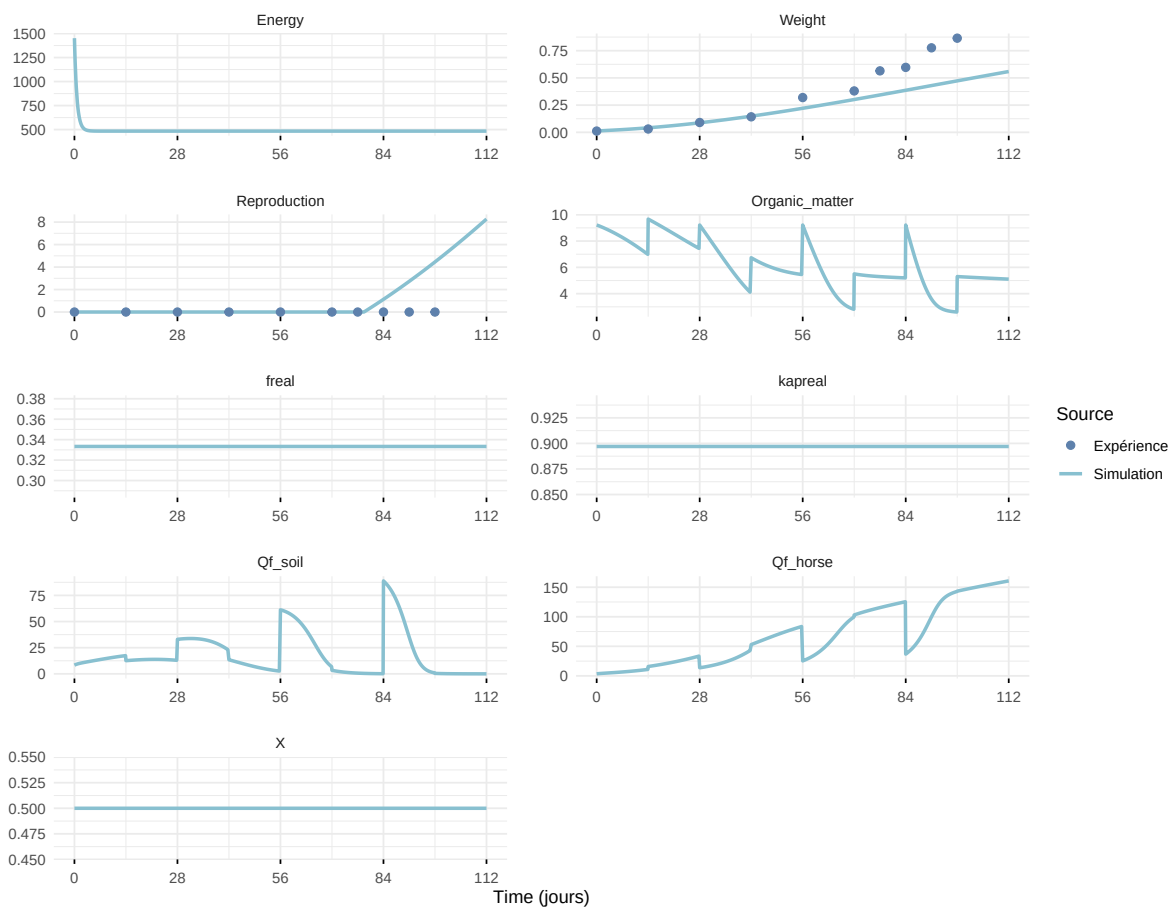
Expérience : Bart2019_XP4_2



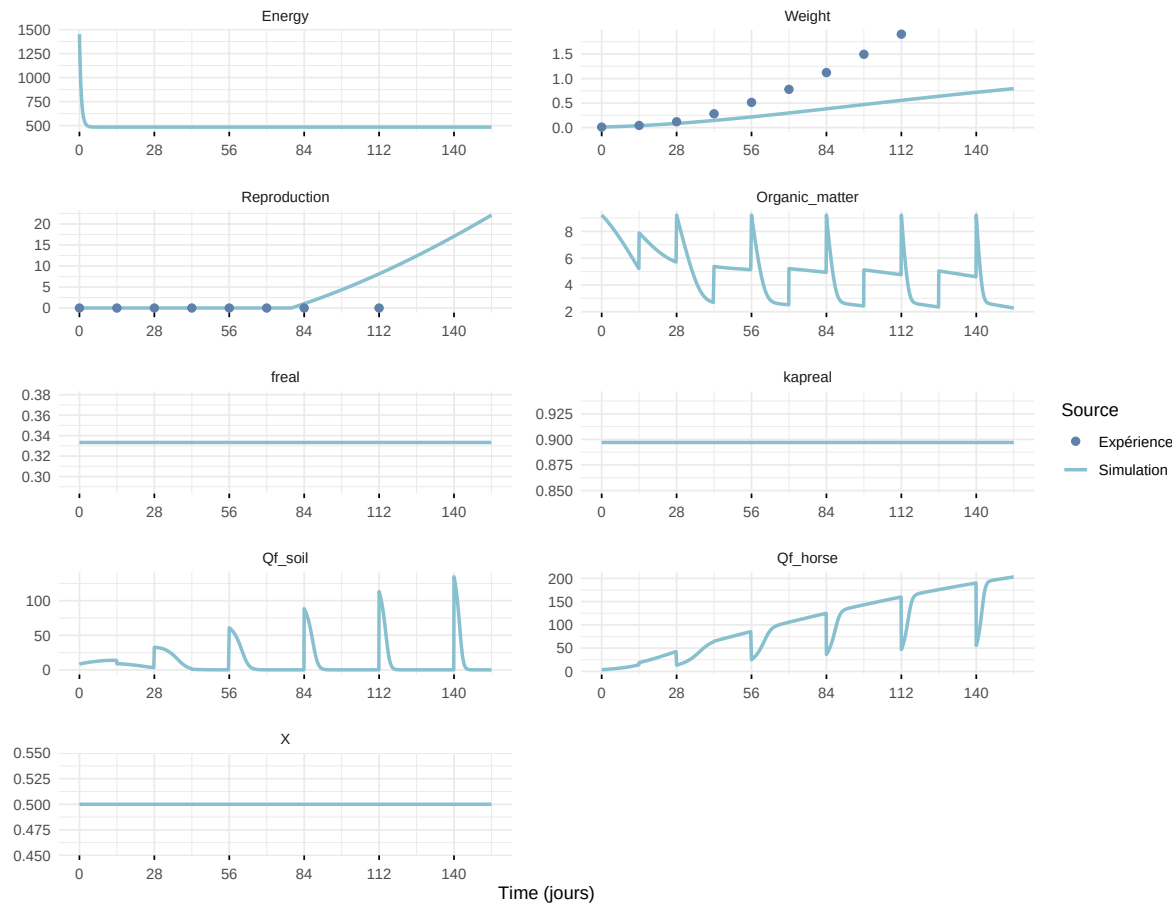
Expérience : Bart2019_XP5



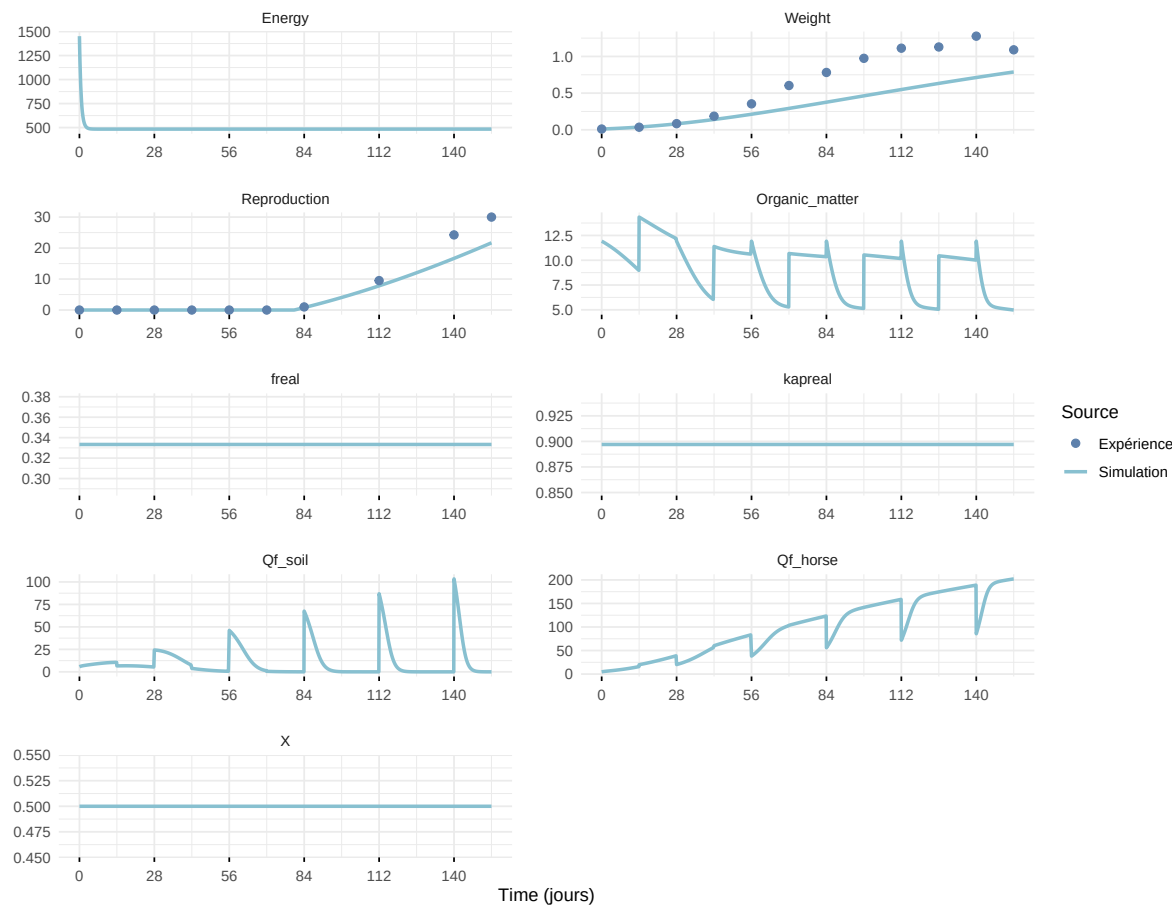
Expérience : Bart2020



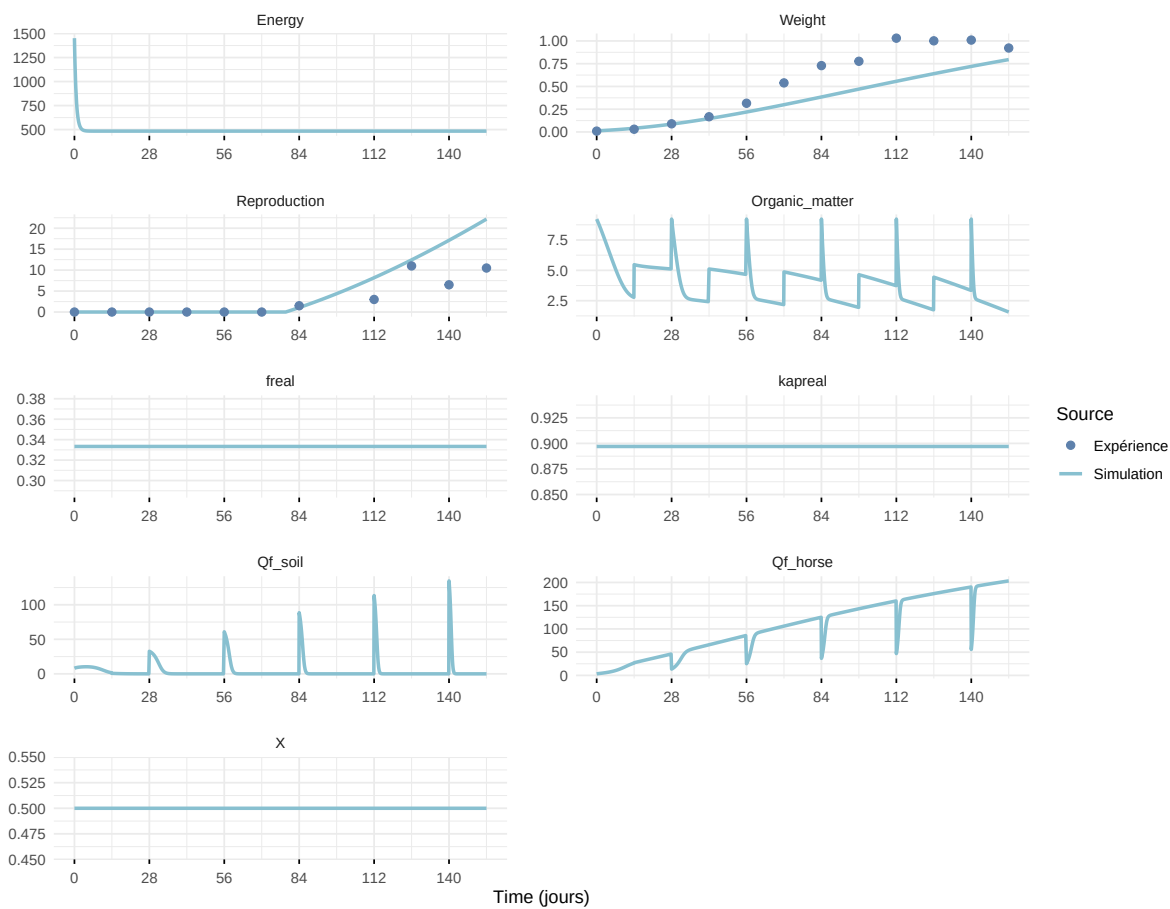
Expérience : Gollot2026_dens_D1



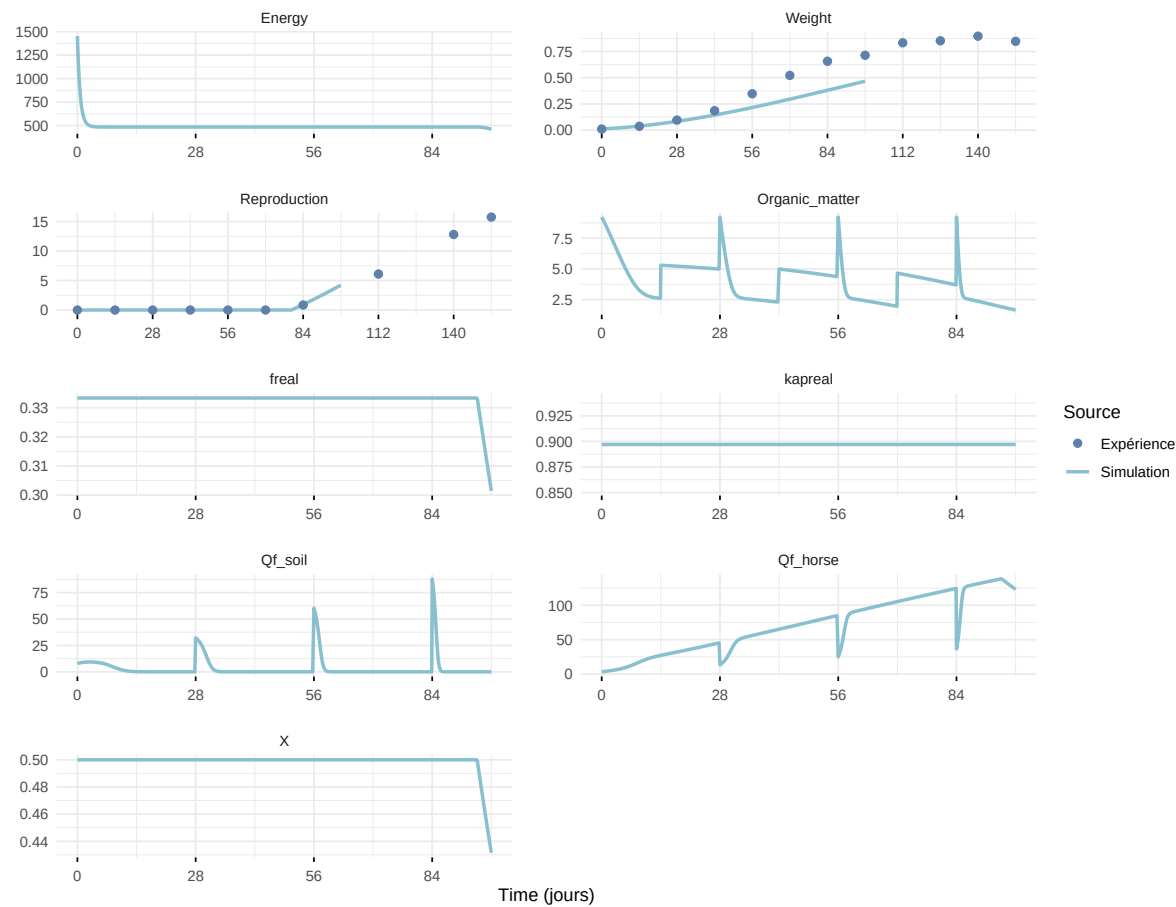
Expérience : Gollot2026_dens_D2



Expérience : Gollot2026_dens_D2b



Expérience : Gollot2026_dens_D5



Expérience : Gollot2026_dens_D7

